Machine Learning Classification of Quantum Particle Energy States

Using a Simulated Image Detector: A Controlled Computational Study

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Abstract – *This study tests whether classical classifiers can identify the first four stationary states of a particle in a one-dimensional infinite well from simulated detector observations. Analytical probability densities generate 1,200 stochastic 32 × 32 images (80 detections each, peak-normalized). k-NN, Decision Tree, Random Forest, and standardized RBF-SVM are trained once on nominal data and tested on held-out observations under reduced resolution, Gaussian blur, and additional noise. Nominal accuracy is 100.0%, 95.8%, 99.2%, and 100.0%, respectively. At 25% resolution and high blur, tree-based accuracy falls to about 46–48%, while SVM and k-NN remain higher; their gap at 25% resolution is within 20-seed variability, and rankings depend on preprocessing. For resolution and blur, retraining on degraded images restores 96–100% accuracy, indicating train/test mismatch rather than lost class information. k-NN and Random Forest remain robust to the tested noise. This is a controlled synthetic robustness benchmark, not a new classification method.*

Keywords: quantum mechanics; particle in a box; machine learning; image classification; detector degradation; robustness; synthetic data

1. Introduction

The one-dimensional infinite potential well is a standard analytical model in which a confined particle has discrete eigenstates with distinct spatial probability-density patterns, making it a controlled setting for pattern recognition.

Machine learning can infer physical states from measurements, but detector resolution and measurement fluctuations can alter information available to a classifier. A controlled synthetic

experiment isolates these effects without the complexity of a real instrument. Such conditions can arise in count-limited, finite-resolution imaging of confined particles.

This study asks whether a classical classifier can identify the first four stationary eigenstates from finite detector observations subjected to controlled degradation. It does not directly measure energy experimentally; the class label is the quantum number n, inferred from a simulated observation generated from $|\psi_n(x)|^2$.

1.1 Hypotheses

Three hypotheses are tested:

H1: All four classifiers will perform substantially above the four-class chance level of 25% on nominal test observations.

H2: Increasing resolution loss and blur will reduce classification accuracy, while increasing additional detector noise may affect classifiers differently and need not produce a monotonic decline for every model.

H3: Robustness will depend on classifier type, so relative model rankings may change across degradation types and random realizations.

2. Background and Related Work

2.1 Quantum particle in a one-dimensional infinite well

For an infinite square well extending from x = 0 to x = L, the normalized stationary wavefunctions are

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right), \qquad n = 1,2,3, \ldots$$

and the corresponding energy eigenvalues are

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}$$

The probability density is $P_{n(x)} = |\psi_{n(x)}|^2$. The first four states therefore have one, two, three, and four probability lobes, respectively. This standard particle-in-a-box formulation follows introductory quantum-mechanics treatment (Blinder, 2021). In this experiment, these patterns—not the numerical value of E_n—are the information presented to the classifier (Figure 1).

2.2 Machine learning and quantum-data classification

Related work includes quantum-state classification from measurement data with uncertainty (Tang et al., 2024), optical quantum-state classification under distortions such as Gaussian noise and photon loss (Ahmed et al., 2021), neural-network energy-level prediction in one-dimensional systems including the particle-in-a-box model (Ogure, 2022; Khalili Golmankhaneh et al., 2026), and controlled corruption tests for image-classifier robustness (Hendrycks & Dietterich, 2019). These studies establish the broader research context.

This study instead evaluates four standard classical classifiers after training on nominal synthetic observations and testing the same models across controlled detector degradations. It does not claim a new algorithm, classifier, or state-classification method; its contribution is a controlled robustness baseline and framework for later studies with more realistic detector physics or experimental data.

3. Methodology

3.1 Quantum data generation

The well length was set to L = 1. The density $P_{n(x)} = |\psi_{n(x)}|^2$ was evaluated at the centres of 32 equally spaced bins, $x_i = \frac{i+0.5}{32}$, and the 32 values were renormalized to sum to one before 80 detections were sampled multinomially. Counts were then divided by each observation's maximum bin count before expansion into a 32 × 32 image. Thus, nominal images retain relative spatial shape but no absolute count scale. Because images are not renormalized after degradation, resolution and blur can also produce class-correlated peak-intensity changes at test time; with peak-normalized training images, this is a distribution shift rather than a learned amplitude cue.

3.2 Simulated detector representation

The one-dimensional 32-bin response is expanded vertically into a 32 × 32 grayscale image; the vertical axis is an artificial representation, not a second physical coordinate. Each image also includes row-wise gain variation (σ = 0.015) and baseline pixel fluctuation (σ = 0.006), which form part of the nominal detector model. The noise experiment adds further Gaussian noise. All settings are listed in Table 2.

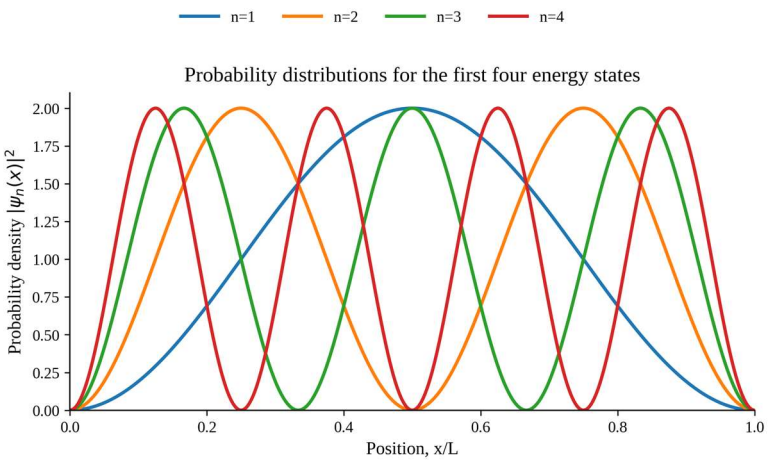


Figure 1. Probability densities of the first four stationary states.

3.3 Detector degradation models

For resolution degradation, images are converted to 8-bit grayscale, reduced to 16 × 16 or 8 × 8, then resized to 32 × 32 using bilinear resampling. The 50% and 25% levels refer to linear resolution; 25% corresponds to 8 × 8 pixels before upsampling, or 6.25% of the original pixel count.

Gaussian blur is applied in 8-bit space with σ = 0.75, 1.50, and 2.50 pixels as low, medium, and high levels; this approximates spatial point-spread degradation rather than a specific detector. Additional Gaussian pixel perturbation uses σ = 0.05, 0.10, and 0.20 and is zero-mean before clipping to [0, 1]. After clipping, it is not strictly zero-mean near intensity bounds; for a zero-valued pixel with σ = 0.20, the expected clipped perturbation is about +0.08. Example degraded observations are shown in Figure 2.

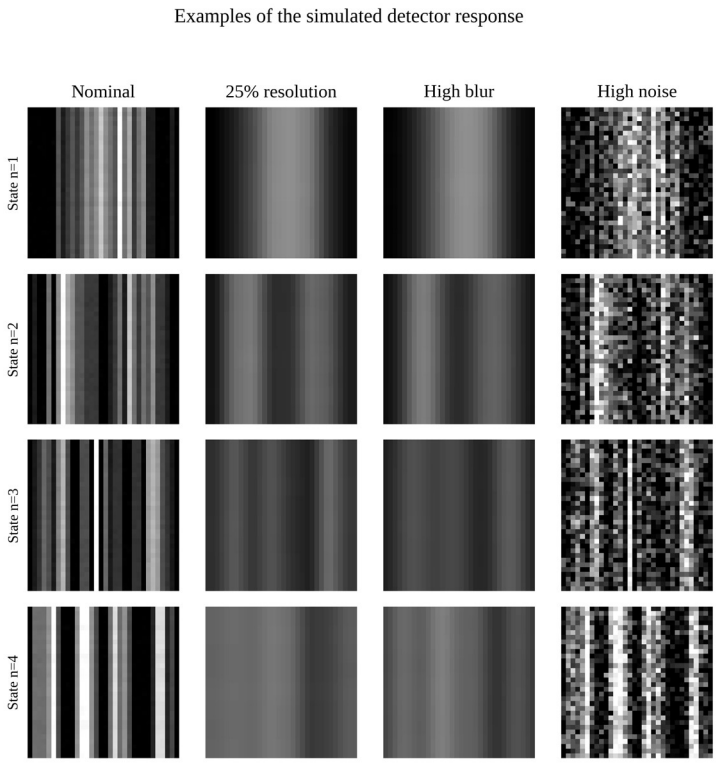


Figure 2. Examples of the simulated detector response for nominal, 25% resolution, high blur, and high noise conditions.

3.4 Train/test protocol

The dataset was split once with a stratified 80/20 split (seed 42) before degradation. Each classifier was fitted once on nominal training observations and evaluated on nominal test observations and degraded versions of the held-out test set. The design measures robustness to a deployment-style train/test distribution shift, not retraining under each degradation; a supplementary retraining check is described in Section 4.6.

3.5 Machine-learning models

Four classical baselines were used: k-NN, Decision Tree, Random Forest, and RBF-SVM. Pixel intensities were flattened into feature vectors; SVM features were standardized. Hyperparameters were fixed before evaluation and not tuned per degradation condition (Table 1).

Classifier	Configuration
k-NN	5 neighbors; distance weighting
Decision Tree	Maximum depth = 14
Random Forest	300 trees; sqrt feature sampling
SVM	RBF kernel; C = 10; gamma = scale; standardized features

Table 1. Fixed classifier configurations.

3.6 Evaluation metric

The primary metric is classification accuracy: the fraction of correct predictions among 240 balanced held-out nominal test images (60 per class). The main results are a fixed-seed benchmark; a secondary 20-seed analysis assesses sensitivity for 25% resolution, high blur, and high additional noise.

3.7 Implementation details

The experiment was implemented in Python. Resolution and blur transformations are applied in 8-bit grayscale space using PIL and converted back to float32 values in [0, 1]; nominal images remain in float32 until a degradation transform is applied. The nominal dataset is generated directly from the analytical quantum model and is not an external dataset. Key settings are summarized in Table 2.

Parameter	Setting
Quantum states	n = 1, 2, 3, 4; density $\|\psi_n(x)\|^2$ on a unit-length well (L = 1)
Dataset size	1,200 images; 300 per state
Detector sampling	80 multinomial detections across 32 bin centres
Count normalization	Each response divided by its maximum bin count (peak = 1)
Detector image size	32 × 32 pixels (32-bin response replicated over 32 rows)
Row-wise gain	Multiplicative factor per image row, $N(1, 0.015^2)$

Parameter	Setting
Baseline pixel fluctuation	Additive N(0, 0.006^2) per pixel, applied after the row gain, then clipped to [0, 1]
Resolution degradation	100%, 50%, 25% linear resolution (32 × 32 → 16 × 16 or 8 × 8 → bilinear resize to 32 × 32)
Blur degradation	Gaussian σ = 0.75, 1.50, 2.50 pixels (8-bit, Pillow)
Additional Gaussian noise	Additive N(0, σ^2), σ = 0.05, 0.10, 0.20, clipped to [0, 1]
Evaluation	One stratified 80/20 split (960 train / 240 test) made before degradation; degradation applied to test data only (except the supplementary retraining check)
Classifiers	k-NN, Decision Tree, Random Forest, RBF-SVM (Table 1)
Model random state	Equal to the run seed
Seeds	Primary 42; sensitivity analysis 100–119 (20 runs)
Software environment	Python 3.13.16; NumPy 2.1.3; pandas 2.2.3; scikit-learn 1.6.1; Pillow 11.3.0; Matplotlib 3.10.0

Table 2. Key dataset, degradation, and evaluation settings.

4. Results

Table 3 and Figure 3 report fixed-seed accuracy on the 240-image nominal test set and on degraded versions of the same test set (four-class chance = 25%).

Condition	Level	k-NN	Decision Tree	Random Forest	SVM
Nominal	—	100.0%	95.8%	99.2%	100.0%
Resolution	50%	100.0%	72.1%	83.8%	100.0%
	25%	76.3%	47.1%	45.8%	81.7%

Condition	Level	k-NN	Decision Tree	Random Forest	SVM
Blur	Low	100.0%	82.1%	98.3%	100.0%
	Medium	100.0%	67.1%	67.1%	100.0%
	High	85.8%	48.3%	46.7%	94.2%
Noise	Low	100.0%	89.6%	99.6%	100.0%
	Medium	100.0%	80.8%	99.6%	100.0%
	High	100.0%	68.8%	99.2%	96.7%

Table 3. Fixed-seed classification accuracy on the nominal test set and under each detector degradation; four-class chance = 25%.

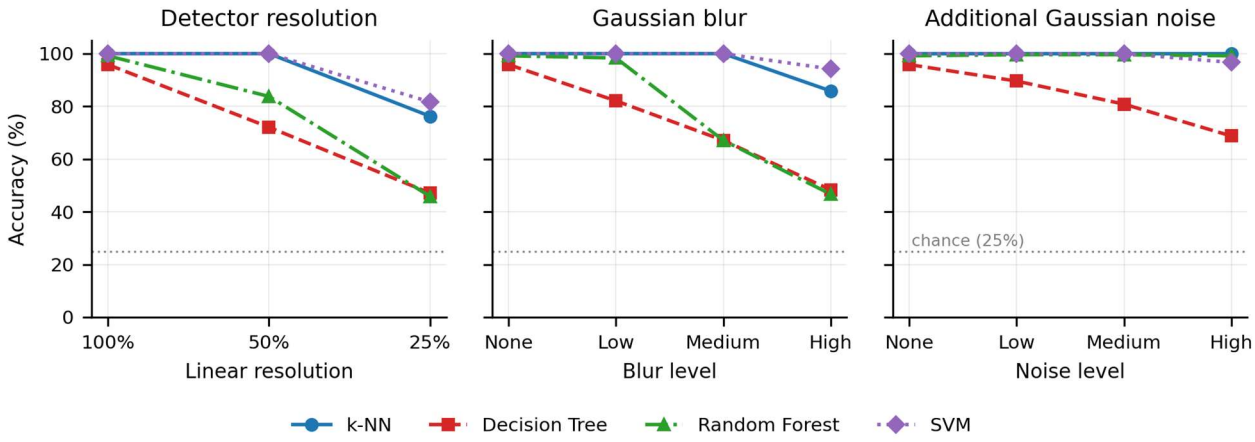


Figure 3. Accuracy versus (left) detector resolution, (center) Gaussian blur, and (right) additional Gaussian pixel noise for models trained on nominal data; the dotted line marks four-class chance.

4.1 Nominal detector observations

k-NN and SVM correctly classified all 240 nominal test observations; Random Forest and Decision Tree misclassified two and ten, respectively. The high accuracy is expected because the four theoretical probability patterns are strongly separated and only four known states are classified.

4.2 Effect of reduced resolution

At 50% resolution, Decision Tree and Random Forest fell to 72.1% and 83.8%, while k-NN and SVM remained at 100%. At 25%, fixed-seed accuracy was 81.7% for SVM, 76.3% for k-NN, 47.1% for Decision Tree, and 45.8% for Random Forest. Across 20 seeds (Table 5), SVM and k-NN averaged 82.4 ± 6.1% and 81.1 ± 2.7%, so the single-seed difference is not a universal ranking.

The tree-based models show a collapse toward n = 1: Random Forest predicts all 60 n = 3 cases and 58/60 n = 4 cases as n = 1 (Figure 4). At 25% linear resolution, the detector representation is reduced to an 8 × 8 image after downsampling. The n = 4 density has spatial period L/4, so only two samples span each period at this resolution. Its pattern can therefore look nearly flat after downsampling, unlike anything in the nominal training set. Because retraining on degraded images restores 95.7–100% accuracy (Table 5), the drop reflects train/test mismatch rather than loss of class-relevant information at this resolution, with spatial and peak-intensity changes both contributing to the shift.

4.3 Effect of blur

High blur reduced Decision Tree and Random Forest accuracy to 48.3% and 46.7%, while k-NN and SVM retained 85.8% and 94.2%. Across 20 seeds (Table 5), mean accuracy was 92.4 ± 4.7% for SVM and 89.3 ± 2.5% for k-NN; tree-based models remained near 48–52%. SVM therefore led under the primary preprocessing convention, but this ranking is conditional on that convention and does not establish a specific learned mechanism.

4.4 Effect of additional Gaussian detector noise

Noise produced a different robustness profile. At the highest tested level, k-NN remained at 100.0%, Random Forest at 99.2%, SVM at 96.7%, and Decision Tree at 68.8%. Across 20 seeds (Table 5), the means were 100.0 ± 0.0%, 99.8 ± 0.3%, 93.1 ± 4.1%, and 69.1 ± 3.4%, respectively. These results are specific to the stated additive-noise model and the standardized RBF-SVM pipeline; they do not imply a universal noise law or robustness of unscaled SVM variants.

4.5 Sensitivity to normalization order

The primary benchmark peak-normalizes nominal images before degradation. In the secondary test, each degraded image is instead peak-normalized after transformation while the fitted models remain unchanged. Table 4 reports all twelve condition–model pairs because normalization order changes some results in both directions.

Classifier	25% resolution		Medium blur		High blur	
	Primary	Post-norm.	Primary	Post-norm.	Primary	Post-norm.
k-NN	76.3%	75.4%	100.0%	100.0%	85.8%	86.7%
Decision Tree	47.1%	27.5%	67.1%	73.3%	48.3%	28.3%
Random Forest	45.8%	53.8%	67.1%	91.3%	46.7%	51.3%
SVM	81.7%	66.3%	100.0%	93.3%	94.2%	73.8%

Table 4. Fixed-seed accuracy with the primary pipeline (peak normalization before degradation) versus post-degradation peak normalization (Post-norm.).

The diagnostic analysis further quantifies the intensity shift. For 25% resolution, mean peak intensity for n = 1–4 is 0.610, 0.556, 0.518, and 0.427; under high blur it is 0.592, 0.538, 0.473, and 0.415. Mean pixel intensity remains about 0.30 across classes. Because training observations are peak-normalized, these test-time differences are a degradation-induced covariate shift, not a learned class-amplitude cue. They arise because the same spatial transformations attenuate patterns with different spatial frequencies, coupling intensity to spatial degradation.

4.6 Repeated-seed sensitivity analysis

Each repetition regenerated the 1,200-image dataset, used a new stratified 80/20 split and new degradation draws, and varied the stochastic tree/forest seeds. Table 3 remains the primary fixed-seed benchmark; Table 5 quantifies sensitivity for three severe conditions and also reports a supplementary check in which each model was retrained on independently degraded copies of the training images and tested on the same degraded test sets.

Condition	k-NN	Decision Tree	Random Forest	SVM
25% resolution	81.1 ± 2.7%	50.6 ± 5.3%	47.0 ± 1.7%	82.4 ± 6.1%
High blur	89.3 ± 2.5%	51.6 ± 4.8%	48.2 ± 1.2%	92.4 ± 4.7%
High additional noise	100.0 ± 0.0%	69.1 ± 3.4%	99.8 ± 0.3%	93.1 ± 4.1%
25% resolution (retrained)	100.0 ± 0.0%	95.7 ± 1.4%	98.8 ± 0.7%	100.0 ± 0.0%
High blur (retrained)	100.0 ± 0.1%	96.2 ± 1.5%	99.0 ± 0.8%	100.0 ± 0.0%

Table 5. Mean ± standard deviation of accuracy over 20 independent seeds for selected severe degradation conditions. Rows marked 'retrained' use models trained on degraded training images.

4.7 Confusion-matrix analysis

Figure 4 shows confusion matrices at 25% resolution and under high blur. At 25% resolution the tree-based models collapse toward n = 1, consistent with train/test mismatch and degradation-induced intensity changes; under high blur, the error patterns reflect classifier-specific sensitivity to the simulated degradation rather than a proven feature mechanism.

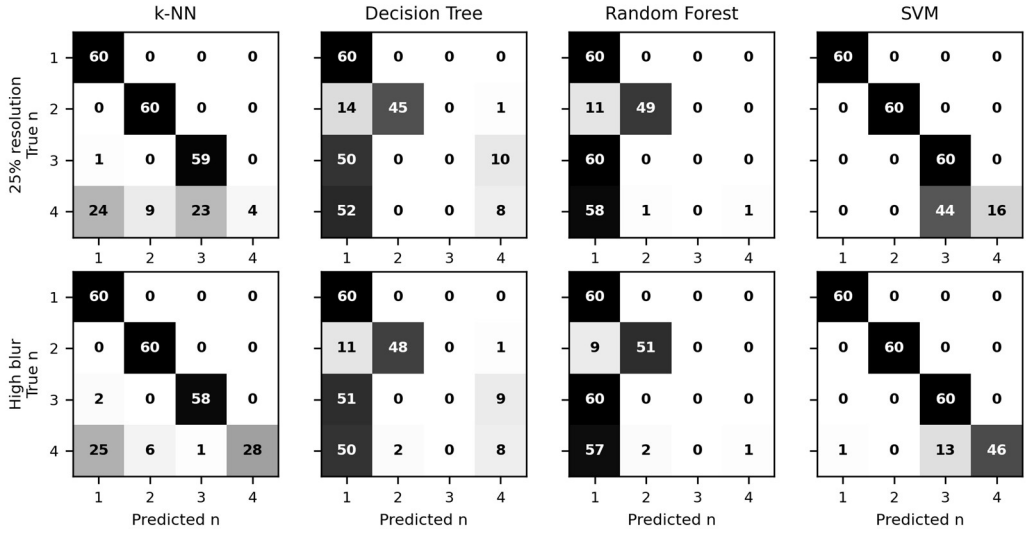


Figure 4. Confusion matrices (true versus predicted n) at 25% linear detector resolution (top row) and under high Gaussian blur (bottom row).

5. Discussion

Three findings stand out. First, under resolution loss and blur, tree-based accuracy falls progressively with severity, and at the most severe levels their errors concentrate on n = 1, whereas k-NN and SVM retain higher accuracy. Second, part of the measured effect is a peak-intensity covariate shift caused by the transformations, so classifier rankings depend on whether images are renormalized after degradation. Third, the strong noise robustness of k-NN and Random Forest may partly reflect the replicated-row image geometry and should not be generalized to real detectors.

The experiments support H1 and H3 and broadly support H2 for resolution and blur. The 20-seed analysis shows that the broad classifier-dependent pattern persists, while small single-seed ranking differences should not be treated as universal.

Resolution and blur alter spatial structure and, because normalization occurs before degradation, can also alter peak intensity, whereas additive noise perturbs pixel values while preserving the large-scale pattern. The strong k-NN and Random Forest results therefore apply only to the specified Gaussian-noise model and geometry. Normalization-order sensitivity (Table 4) confirms that preprocessing affects observed robustness: the primary benchmark is a fixed detector-processing convention, not a pure test of spatial-shape robustness.

Training once on nominal data isolates deployment-style train/test mismatch. For 25% resolution and high blur, retraining on degraded images restored 95.7–100% accuracy (Table 5), so those failures reflect distribution shift rather than loss of class-relevant information; this was not tested for noise. The classifiers infer the discrete state label from a known, highly structured simulated distribution, so the value of the experiment lies in the controlled robustness comparison, not in a new quantum-energy measurement method.

6. Limitations and Future Work

- **Synthetic detector and noise model:** the simulation omits physical effects such as nonuniform response, calibration drift, dead pixels, realistic point-spread functions, electronic readout noise, background counts, saturation, correlated noise, and alignment errors, and additive Gaussian noise is only one possible noise model.
- **Artificial geometry and replicated-row noise averaging:** the vertical dimension is a replicated one-dimensional response, not a second physical coordinate. Repeated rows can also let models effectively average independent pixel noise, which may contribute to strong robustness observed under the tested pixel-noise model and limits generalization to real detectors.
- **Restricted state space and fixed settings:** only four analytically known states are used, so the task tests classification among fixed classes rather than generalization to new physical systems; hyperparameters were specified in advance rather than optimized, and should not be interpreted as globally optimal.
- **Single primary split:** the main benchmark uses one stratified 80/20 split. The secondary 20-seed analysis quantifies variability for three severe conditions but does not replace repeated evaluation across all conditions.

Future work could vary the number of states, the number of detections, and the detector bin width relative to the well; extend the study to other potentials (e.g., finite wells or harmonic oscillators); introduce physical point-spread and counting-noise models; extend repeated-seed evaluation to all conditions; report confidence intervals and macro-F1; and compare the baselines with convolutional models.

7. Conclusion

This study provides a controlled computational experiment for classifying the first four stationary eigenstates of a particle in a one-dimensional infinite well from synthetic detector observations. Analytical densities were converted into stochastic grayscale images, four classifiers were trained once on nominal observations, and the same models were evaluated under controlled resolution, blur, and additional Gaussian-noise degradation.

Detector degradation affected classifiers differently. Under the primary preprocessing convention, severe resolution loss and blur strongly reduced tree-based accuracy, while SVM and k-NN retained higher performance. The 20-seed analysis shows that their average accuracies at 25% resolution are close, and normalization order can change rankings. k-NN and Random Forest remained highly accurate under the tested additional Gaussian noise.

The study is a controlled synthetic robustness benchmark, not a real-detector demonstration or a new quantum-state-classification algorithm. Its principal strength is experimental control: the analytical model, stochastic sampling, degradation settings, seeds, split, and classifier configurations are specified in enough detail to be re-implemented. Code is available from the author on request.

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